

3D FE simulation for temperature evolution in the selective laser sintering process

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Abstract

A thermal model of selective laser sintering (SLS) has been developed. The model allows for the non-linear behavior of thermal conductivity and of specific heat due to temperature changes and phase transformations. The temperature evolution and the formation of the sintered part are simulated by a 3D finite element analysis based on continuous media theory. It is shown that the effect of sintering has a strong influence on thermal evolution through changing the thermal properties of the material. The results of the model were experimentally tested and confirmed by temperature measurements.

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1. Introduction

Selective laser sintering (SLS) is a solid freeform fabrication technique. It consists of building a three-dimensional object layer by layer out of a powder selectively fused by laser radiation. Under the laser radiation, the powder partially melts. The liquid formed by the molten material binds the surrounding powder and solidifies when the temperature decreases, which leads to consolidation. The presence of the liquid for a short time leads to a shrinkage of the powder–liquid mixture. Thus, the density of the material depends on the temperature evolution. The consolidated structure is a rigid and porous object. The porosity cannot be completely eliminated, but the density of the manufactured part is usually higher than the powder density. Moreover, due to temperature variation, mechanical stresses are induced. As a consequence, the final state of the part (dimensions, density,

residual stress levels, etc.) strongly depends on the process evolution.

In most reported works on the simulation of SLS, the model of the process is usually limited to one or two space dimensions. In Refs. [7,10,14], various hypotheses of symmetry are assumed and the space dimension is two. To get pseudo-stationary equations similar to Rosenthal's equation, the authors of [3] limit their analysis to an extremely simple scanning strategy. In real sintering processes, the laser strategy is a complicated broken line and usually the process has no translational or planar symmetry. These two facts prohibit reduction of the dimension of the problem.

The descriptions of the SLS process are usually based on the hypothesis of constant thermal conductivity. Only some works allow for the evolution of thermal conductivity. The model of Yagui-Kunni is used in [3,9,10]. It assumes the thermal conductivity to be in direct proportion to the solid fraction (or density in other words). As has been observed before, the density of sintered material can vary from the density of loose powder (about 50% of the solid density) up to the solid density (never reached in sintering). Thus, the ratio

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$\rho_{\max}/\rho_{\min}$ is less than 2. However, the thermal conductivity grows from the value of loose powder (only a small percentage of solid thermal conductivity for metallic powders) up to the thermal conductivity of the solid. Thus, the ratio of maximum to minimum thermal conductivity can vary from 10 to 50: $k_{\max}/k_{\min} \sim 10\text{--}50$. To summarize, the thermal conductivity growth cannot be explained only by the effect of grain compaction. Metallurgical bonding between grains also takes place in sintering, which leads to the formation of good heat-conductive links.

A fully 3D thermal model of SLS describing the non-linear effects of sintering has been developed. This work focuses on a model for the evolution of thermal conductivity. It is known that the thermal characteristics of the bulk material, such as thermal conductivity and heat capacity, are temperature dependent. Moreover, during sintering, the thermal conductivity and the density of the powder also depend on the thermal history, since the workpiece structure evolves with time.

The paper is organized as follows: Section 2 presents the model of the SLS process described above. Section 3 presents the numerical algorithm. A description of the numerical and physical experiments can be found in Section 4. The results are presented in Section 5 and conclusions follow in Section 6.

2. Modelling approach

2.1. Concept

The physical model of the SLS process is based on the continuous media theory. The fields describing the evolution of the medium are defined in the domain Ω , where the SLS process is localized. These fields are the temperature T , the volume enthalpy u and the sintering potential ϕ . The sintering potential ϕ is defined in point $\mathbf{x}$ at time t as follows:

$$\phi(\mathbf{x}, t) = 1 - \exp\left(-\int_0^t \zeta(T(\mathbf{x}, s)) \, ds\right), \quad (1)$$

where $\zeta(T)$ represents the “sintering rate” and is temperature dependent. A model by Frenkel [6] expresses the sintering rate in terms of the surface tension γ and the viscosity μ of the molten material:

$$\zeta(T) = \frac{\gamma(T)}{\mu(T)}.$$

The sintering potential ϕ contains information about the sintering state of the material. The range of ϕ is

$[0, 1)$. The value $\phi = 0$ corresponds to loose powder, whereas ϕ tends to 1 in fully sintered material.

2.2. Laser radiation

In the model presented here, the laser radiation is considered as a heat source. The geometry of this source is defined by the absorption phenomena in the skin layer. The laser radiation can penetrate into the solid but only up to a certain distance from the surface. This distance is called “penetration depth”. It depends on the material properties, such as permittivity and density. In the case of bulk metals, the penetration depth is very small and is about one part of the wavelength of the incident electro-magnetic wave. The Nd:YAG laser has a wavelength of 1.064 μm . Consequently, the penetration depth in a solid metal is less than 1 μm . For bulk titanium, the penetration depth is 6 nm. In porous materials, the penetration depth is higher than in bulk material. It also depends on the structure of the object. For monosized powders, the penetration depth can reach up to several particle diameters [2,12]. Usually the maximum size of the powder grains is known and the upper limit of the penetration depth might be computed. Observe however, that it is in general difficult to estimate the penetration depth precisely. It strongly depends on the material properties and can range (for similar granulometry) from 20 μm (nickel) up to 63 μm (titanium) [11]. Anyway, if the finite element (FE) elementary size is larger than five grain diameters, then the laser heat source can be considered as a surface heat source.

2.3. Thermal analysis

In the case considered here, the analysis of the SLS process leads to a three-dimensional, non-linear heat transfer problem without volume heat source. The temperature T and the enthalpy u come from solutions to the heat equation:

$$\frac{\partial u}{\partial t} - \text{div}(k \nabla T) = 0, \quad \text{in } \Omega, \quad \text{for any } t > 0, \quad (2)$$

where k denotes the thermal conductivity. It depends on the temperature and the sintering potential. If $k_{\text{bulk}}(T)$ is the thermal conductivity of the bulk, one can write:

$$k = (a + (b - a)\phi)k_{\text{bulk}}(T), \quad (3)$$

where $a > 0$ is the ratio between the conductivity of loose powder and the bulk conductivity and $b > 0$ is the ratio between the conductivity of completely sintered material and the bulk conductivity. These are assumed

to be temperature independent. Since a porous material never conducts heat as well as bulk material (see Section 1), we will have $a < 1$, $b < 1$. Typical values are: $a = 0.01$, $b = 0.6$.

Eq. (2) goes together with a condition on the boundary $\partial\Omega$, which allows for laser heating and surface cooling, due to thermal exchange with the surroundings. It reads

$$-k \frac{\partial T}{\partial n} = \alpha(T^{\text{inf}} - T) + F, \quad \text{on } \partial\Omega, \quad \text{for any } t > 0, \quad (4)$$

where α is the thermal exchange coefficient between the domain Ω and the environment, T^{inf} is the temperature of the environment and F is the boundary source. It represents laser heating. Observe that $\partial T / \partial n$ denotes the outward normal derivative of the temperature on the surface $\partial\Omega$.

The model is completed by a state law linking the volume enthalpy to the temperature:

$$T = \beta(u). \quad (5)$$

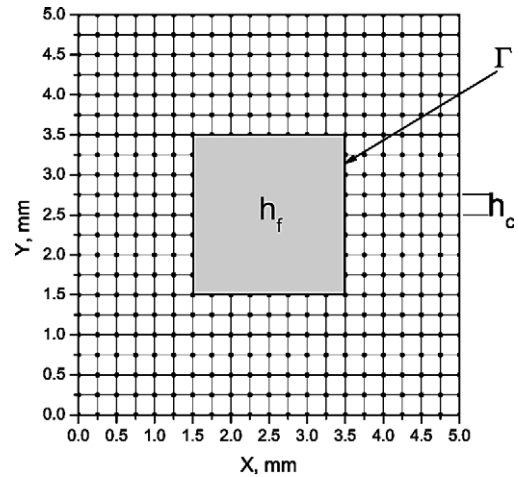
In Eq. (5), β is a non-decreasing function and $(\beta')^{-1}$ can be interpreted as the specific heat C_p .

3. Numerical algorithm

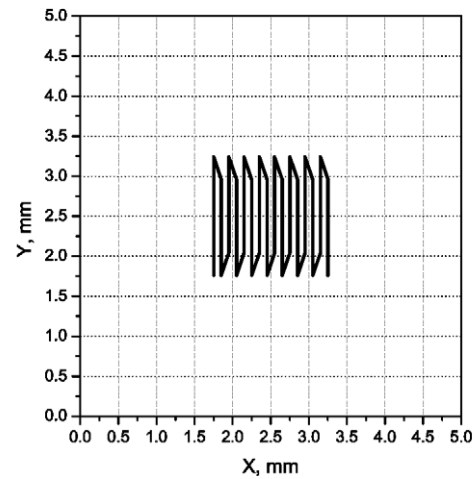
A finite element method (FEM) for space discretization combined with a Chernoff scheme for time discretization has been chosen to solve (2), (4), (5). It has been proved in [8] that this method provides a fully discrete scheme converging to the solution (u, T) to problem (2)–(5) as soon as the time step and the mesh size tend to zero.

The FE mesh used in this work is a quasi-regular mesh with two types of cells (see Fig. 1(a)), coarse cells of typical size h_c and fine cells of typical size h_f . The associated FE method is non-conforming along the interface Γ between the coarsely and finely meshed regions [1].

In comparison, the conformal meshes used in Refs. [4,5] are much more difficult to build. Observe that the zone with fine cells has to cover, at least, the area of laser heating and some surroundings. The values h_c and h_f have to be chosen in such a way as to diminish the error of computation to an appropriate level and to avoid irresolvable growth in computational complexity. The upper limit of h_f will anyway be less than the size of the laser spot.



(a) The mesh view.



(b) The laser path.

Fig. 1. View from above of the quasi-regular mesh and of the laser path.

4. Experiments

Both numerical and physical experiments have been performed. The physical parameters in these experiments are similar and presented below.

4.1. Laser characteristics

Variable	Notation	Value	Unit
Average power	P	2.0	[W]
Spot radius	R	2.5×10^{-5}	[m]
Laser spot speed	S	1.0×10^{-3}	[m/s]

The laser spot moves on the top surface of the piece with speed S . The laser operates in continuous wave mode (CW) and the laser path is presented in Fig. 1(b).

4.2. Powder characteristics

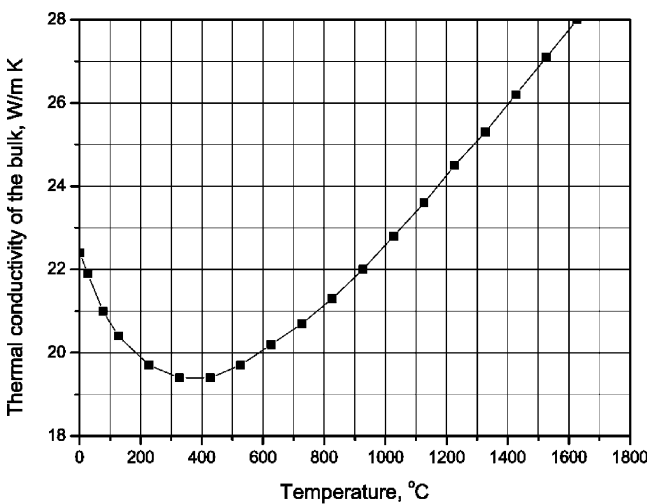
The parameters of the material are given in the following table:

Variable	Notation	Value	Unit
Density	ρ	2.9×10^3	[kg/m ³]
Specific heat (average)	$\langle C_p \rangle$	1.9×10^{-3}	[J/mm ³ K]

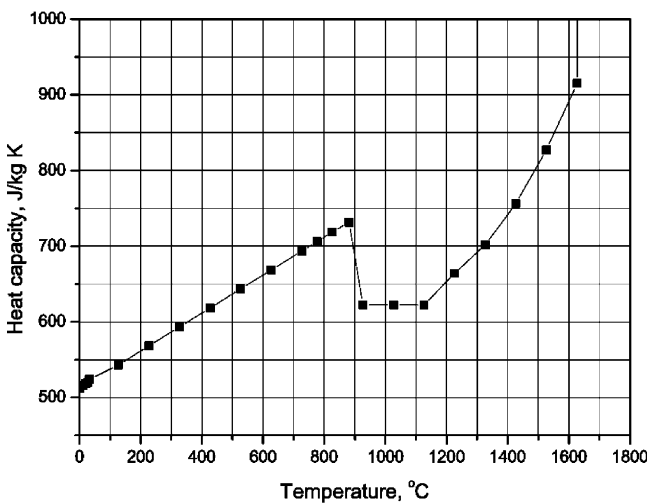
The dependence of the thermal conductivity $k_{\text{bulk}}(T)$ and specific heat $C_p(T)$ on temperature are presented in Fig. 2(a), (b) and they are taken from handbook [13].

4.3. Numerical experiment

The dimensions of the computational domain are $5 \times 5 \times 2$ mm. The sizes of the coarse cells h_c and of



(a) Thermal conductivity.



(b) Specific heat.

Fig. 2. Thermodynamic properties of solid titanium. (a) Thermal conductivity. (b) Specific heat.

the fine cells h_f are 0.1 and 0.01 mm, respectively. The fine mesh covers the region $2 \times 2 \times 0.5$ mm in the center of the domain. All the external faces have a natural boundary condition with zero flux except the bottom face, which has a condition of perfect cooling, the external temperature being 20 °C. The integration time step is 0.05 s. The time is sufficiently small to avoid laser power modulation due to the numerical scheme. The density of the material is assumed to be constant and the specific heat is a temperature dependent function only (see Fig. 2(b)).

Since data on molten metals are very poor, we propose a piecewise linear model for the sintering rate $\zeta(T)$:

$$\zeta(T) = \begin{cases} 0, & T \leq T_{\text{sintering}}, \\ (T - T_{\text{sintering}})/180, & T > T_{\text{sintering}}, \end{cases} \quad (6)$$

where $T_{\text{sintering}} = 1300$ °C represents the typical sintering temperature. Eq. (6) means that the sintering process only starts above a threshold. One can assume that the powder, heated slightly below melting temperature for some seconds, results in either well-sintered or bulk-like material. The coefficient 1/180 is a calibration parameter and is chosen in such a way that the sintering potential ϕ reaches the value 0.86 (well-sintered state) if the temperature of melting $T_{\text{melt}} = 1660$ °C is maintained for 1 s. Moreover, the value of 1/180 gives the best fit of the simulated temperature profiles to the experimental measurements.

4.4. Physical experiment

A titanium power¹ layer of 2 mm thickness was deposited on a steel substrate without any pre-treatment. The Nd:YAG laser in CW mode scanned the surface according to the path presented in Fig. 1(b) (the distance between each scan line is 0.1 mm). The temperature field on the top of the powder layer was filmed with an infra-red camera² with a spectral band filter from 3 up to 5 μm (see Appendix A). The visible square has 2.8×2.8 mm size in focus. The resolution of the camera is 256×256 pixels. The picture sequence was filmed with a frame rate of 10 Hz. The distance that the laser can cover during the integration time of the camera (0.01 s) is about 0.01 mm or 1 pixel.

5. Results and discussion

5.1. Comparisons

Fig. 3(a), (b) shows the temperature profiles predicted by simulation (bold) and measured by the infra-

¹ PyroGenesis: CPTi powder, Grade 2, size: up to 25 μm .

² Raytheon: Radiance HS Camera, serial number: 2037.

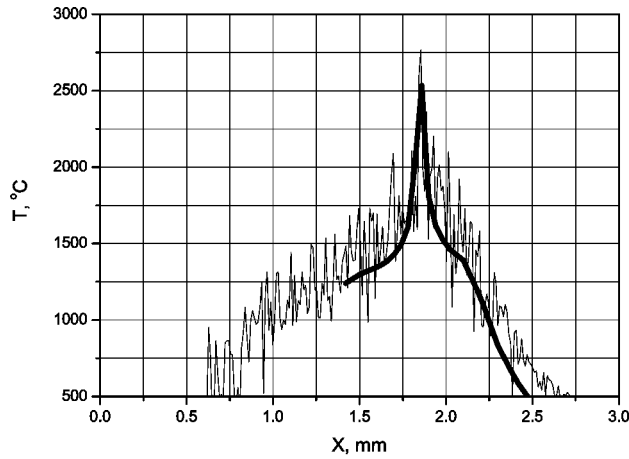
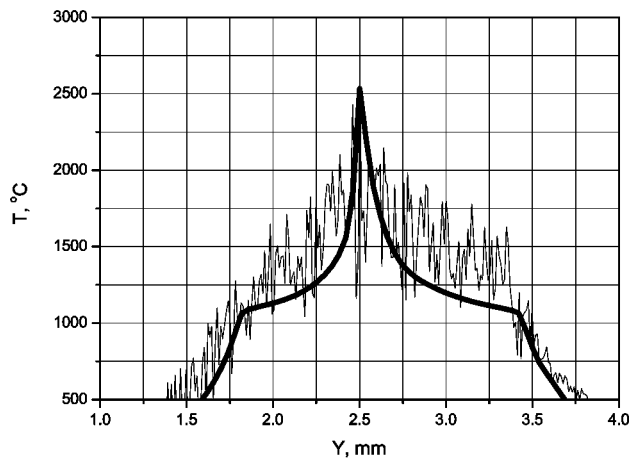
(a) X-profile, $Y=2.5$, $Z=2.0$.(b) Y-profile, $X=1.85$, $Z=2.0$.

Fig. 3. Simulated (bold) and experimental (thin) temperature profiles. (a) X-profile, $Y=2.5$, $Z=2.0$. (b) Y-profile, $X=1.85$, $Z=2.0$.

red camera when the laser is in the middle of the long stripe (see Fig. 1(b)) at the point with coordinates $X = 1.85$, $Y = 2.5$, $Z = 2.0$. Observe that the temperatures less than 500°C are not presented. The camera data are not reliable below this threshold.

Fig. 3 demonstrates that the simulated temperature profiles agree with the experimental results:

1. The prediction for the maximum value of the temperature under the laser beam is 2533°C , whereas the measurement is $2400 \pm 200^\circ\text{C}$.
2. Our model allows for the evolution of the thermal conductivity. As soon as the temperature exceeds the threshold of 1300°C , the conductivity grows quickly. On the other hand, the conductivity remains low in the regions with lower temperature. The jump of the thermal conductivity can only be compensated by a jump of temperature gradients to ensure heat flux continuity (there is no phase transition at 1300°C , see Fig. 2(b)). As can be seen in Fig. 3(b), both the

measured and simulated profiles reproduce this phenomenon (at $Y_1 = 1.74\text{ mm}$ and at $Y_2 = 3.42\text{ mm}$).

5.2. The three-dimensional model

It can be seen in Fig. 3(a), (b) that the temperature profiles, solution to the thermal problem (2)–(5), have in general no spatial symmetry. The temperatures also vary with depth z from 20°C at the bottom to 2533°C under the laser beam. Consequently, any reduction in the dimension of the model would lead to non-realistic predictions of the temperature field evolution and a three-dimensional approach is necessary to get accurate results.

5.3. Updating thermal conductivity

Allowing for the dependence of thermal conductivity on temperature and on sintering potential complicates the model a lot and raises the computational cost by a factor of about two. To save time, the simulation of the sintering experiment with a constant thermal conductivity model was tried. Its value can be chosen in the range between the thermal conductivity of the powder ($k_{\text{powder}} = 0.2\text{ W/m K}$) and of the bulk ($k_{\text{bulk}} = 22\text{ W/m K}$). With the average value $k = 11\text{ W/m K}$, the temperature profiles presented in Fig. 4(a), (b) were obtained. The bold lines correspond again to simulation and the thin lines to experimental measurements.

It is clear that a model with constant thermal conductivity fails to reproduce the real temperature pattern. Fig. 4(a), (b) demonstrates that the temperature evolution is strongly influenced by the thermal conductivity. They also prove that this very sensitive parameter has to be computed by taking the thermal history into account.

6. Conclusion

A three-dimensional heat transfer model of the SLS process has been developed. The results of the model are presented and validated experimentally. It has been shown that the temperature field obtained with the model calculations fits well with the measured temperature field upon the interaction of a Nd:YAG laser with a titanium powder bed. The phenomena observed in this work indicate that the evolution of thermal conductivity determines the behavior and development of the thermal processes. Corrections of the model for the case of liquid state sintering are to be done.

The growth in simulation complexity is partially compensated by a technique named “local refinement” for FE models. In combination with quasi-regular mesh construction, this technique allows a large reduction in computation time.

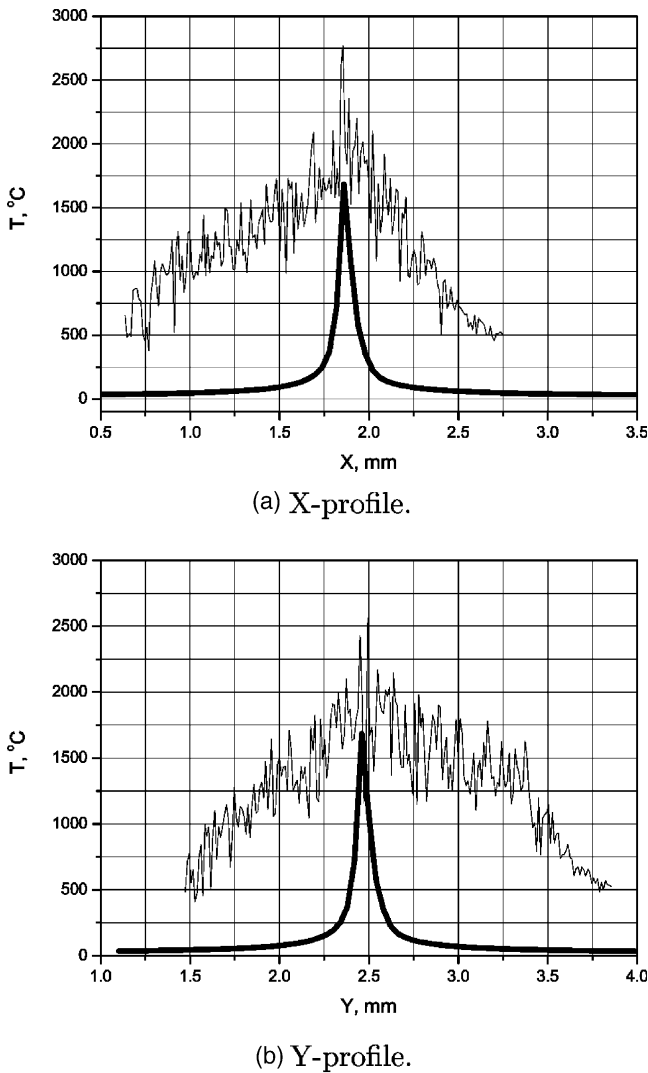


Fig. 4. Simulated (bold) and experimental (thin) temperature profiles. (a) X-profile. (b) Y-profile.

The next goals of the model development are to take the density evolution and residual stresses into account. Further refinement of the model should provide the necessary insight for a better understanding of the sintering phenomena.

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Appendix A. Thermal properties measurement

The infra-red camera allows one to obtain the temperature distribution on the irradiated surface. Some physical properties of the material can be extracted by comparing the distribution of the energy source with the resulting temperature field. The simplest analysis can be performed assuming that the properties being sought are independent of temperature and time. In the case of non-moving laser with a stepwise energy distribution, the thermal conductivity can be expressed analytically as a function of the maximum temperature $T_{\max}$ of the radius R of the source and of the intensity I :

$$k = \frac{IR}{T_{\max}}.$$

This formula follows from Rosenthal's equation. The thermal conductivity can also be found if the source has a more complicated distribution. In this case, the formula for thermal conductivity is:

$$k = c \frac{IR}{T_{\max}},$$

where c is a correction coefficient. The values of the correction coefficient for a gaussian distribution and for a cone source distribution, respectively, are:

$$c_{\text{gaussian}} \simeq 1.27,$$

$$c_{\text{cone}} \simeq 1.75.$$

References

- [1] E. Boillat, Finite element methods on non-conforming grids by penalizing the matching constraint, *M2AN* 37 (2) (2003) 357–372.
- [2] P. Fischer, N. Karapatis, V. Romano, R. Glardon, H.P. Weber, A model for the interaction of near-infrared laser pulses with metal powders in selective laser sintering, *Applied Physics A* 74 (1) (2002) 467–474.
- [3] G. Bugada, M. Cervera, G. Lombera, Numerical prediction of temperature and density distributions in selective laser sintering process, *Rapid Prototyping Journal* 5 (1) (1999) 21–26.
- [4] G. Chen, X. Xu, Experimental and 3D finite element studies of CW laser forming of thin stainless steel sheets, *Journal of Manufacturing Science and Engineering* 123 (2) (2001) 66–73.
- [5] T. Mackawa, G. Yu, K. Masubuchi, N.M. Patrikalakis, FEM simulation of laser forming of metal plates, *Journal of Manufacturing Science and Engineering* 123 (8) (2001) 405–410.
- [6] J. Frenkel, Viscous flow of crystalline bodies under the action of surface tension, *Journal of Physics USSR* 9 (1945) 385–391.
- [7] M. Matsumoto, M. Shiomi, K. Osakada, F. Abe, Finite element analysis of single layer forming on metallic powder bed in rapid prototyping by selective laser processing, *International Journal of Machine Tools and Manufacture* 42 (2001) 61–67.
- [8] E. Magenes, R.H. Nochetto, C. Verdi, Energy error estimates for linear scheme to approximate nonlinear parabolic problems, *Mathematical Modelling and Numerical Analysis* 21 (4) (1987) 655–678.

- [9] M. Kandis, T.L. Bergmann, A simulation-based correlation of the density and thermal conductivity of objects produced by laser sintering of polymer powder, *Journal of Manufacturing Science and Engineering* 122 (2000) 439–444.
- [10] J.C. Nelson, S. Xue, J.W. Barlow, J.J. Beaman, H.L. Marcus, D.L. Bourell, Model of the selective laser sintering of bisphenol—a polycarbonate, *Industrial Engineering and Chemical Research* 32 (1993) 2305–2317.
- [11] P. Fischer, V. Romano, H.P. Weber, Near infrared pulsed laser sintering of metals powder, Technical report, Internal report, University of Bern.
- [12] R. Glardon, N. Karapatis, V. Romano, Influence of Nd:YAG parameters on the selective laser sintering of metallic powders, *Annals of the CIRP* 50 (2001) 133–136.
- [13] Y.S. Touloukian, C.Y. Ho, *Thermophysical Properties of Matter*, IFI/Plenum. New York-Washington. The TPRC Data series, 1970.
- [14] N.K. Vail, B. Balasubramanian, J.W. Barlow, H.L. Marcus, A thermal model of polymer degradation during selective laser sintering of polymer coated ceramic powder, *Rapid Prototyping Journal* 2 (3) (1996) 24–40.